

Windows Security Monitoring

From Logging Fundamentals to Full Threat Detection

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Phase 1	Phase 2	Phase 3	Phase 4
<i>Foundations and Log Sources</i>	<i>Initial Access Detection</i>	<i>Post-Breach Activity</i>	<i>C2, Persistence, and Impact</i>

Project Overview

This document covers the Windows Security Monitoring learning path on TryHackMe, a four-part series covering how Windows-based attacks are detected at every stage using log evidence.

The path is structured progressively. It begins with foundational knowledge of Windows log sources, then moves through each phase of a real attack chain: Initial Access, post-breach activity, command and control, persistence, and impact.

All four labs use real Windows event log files (evtx) and Sysmon logs, analyzed through Windows Event Viewer. No commercial SIEM tool is required. The focus is on building raw log analysis skills that all higher-level detection work depends on.

Project Objectives

- Understand the Windows event logging system, locate log files, and navigate Event Viewer to filter and read events.
- Learn key Windows Event IDs used in threat detection: 4624, 4625, 4688, 4720, 4728, 4698, and Sysmon events 1, 3, 11, and 13.
- Understand Sysmon and why it provides greater visibility than default Windows logs, covering process creation, network connections, file drops, DNS queries, and registry changes.
- Detect Initial Access techniques including RDP brute-force, phishing-delivered malware, and malicious USB execution.
- Investigate post-breach commands, credential harvesting, and sensitive file discovery using Sysmon process logs.
- Detect C2 communication, persistence via registry Run keys, scheduled tasks, and backdoor account creation.
- Recognize the Impact phase when an attacker stops hiding and begins causing damage such as ransomware deployment.
- Map all detected activity to MITRE ATT&CK, connecting log evidence to real adversary techniques.

Tools Used

Tool	Purpose in This Lab
Windows Event Viewer	Primary log analysis interface. Used to open .evtx files, apply Event ID filters, and read individual log entries for authentication, process, and user activity.
Sysmon (System Monitor)	Free Microsoft Sysinternals tool that enriches Windows logging with detailed process creation (Event 1), network connections (Event 3), file drops (Event 11), and registry changes (Event 13).

Windows Security Log (.evtx)	Records authentication events (4624, 4625), user management (4720, 4728), and scheduled task creation (4698). The primary log for detecting credential-based attacks.
PowerShell Logs	Captures all PowerShell commands executed on the machine. Used to detect script-based attacker activity that would not appear in standard process logs.
Event Viewer Filter	Built-in Event Viewer feature to filter log entries by Event ID, time range, and computer name. Used in every phase to isolate specific event types from large log files.
Logon ID Correlation	A session identifier embedded in Windows events. Used to pivot from a login event (4624) to all subsequent actions performed in that session, linking attacker commands to their entry point.

Investigation Methodology

All investigations in this lab followed a consistent four-step methodology applied to each phase of the attack chain.

Step	Action	Description
1	Triage	Open the relevant .evtx log file and apply an Event ID filter to surface only the event type being investigated, removing noise from unrelated entries.
2	Filter and Extract	Read individual matching events to extract key fields: source IP, username, Logon ID, process name, file path, and timestamp.
3	Correlate	Use shared identifiers such as Logon ID or process GUID to link events across different log sources, building a connected sequence of attacker actions.
4	Map and document	Assign each confirmed activity to a MITRE ATT&CK technique and record the evidence in an investigation report covering Who, What, When, Where, and Impact.

Phase 1: Foundations

Windows continuously generates logs for almost every system action. These logs are stored as .evtx files and accessed through Event Viewer. This phase establishes what each log source records and which Event IDs are critical for security investigation.

Critical Event IDs

Event ID	Log Source	What It Records
4624	Security	Successful login. Logon Type 10 specifically identifies an RDP session.
4625	Security	Failed login attempt. High frequency from one IP indicates brute force.
4688, Sysmon 1	Security, Sysmon	New process has started. Sysmon Event 1 includes the full command line and parent process name.
4720	Security	New user account created. A critical indicator of attacker persistence via backdoor account.
4728, 4732	Security	User added to a privileged group such as Administrators or Remote Desktop Users.
4698	Security	Scheduled task created. A common persistence technique used to survive reboots.
Sysmon 3	Sysmon	Network connection initiated by a process. Used to detect C2 communication.
Sysmon 11	Sysmon	File created on disk. Used to detect malware being dropped by a parent process.

Hands-On Exercise

A practice Security log file (Practice-Security.evtx) was opened and filtered for Event ID 4625. A single external IP was identified as the brute-force source. Filtering for 4624 with Logon Type 10 confirmed the RDP session that breached the machine.

The Logon ID from the successful login was then used in Sysmon logs to pivot and identify every command the attacker ran after gaining access, linking the login event to post-compromise activity.

Practice-Security Number of events: 84

Level	Date and Time	Source	Event ID	Task C...
Information	5/17/2025 10:56:18 PM	Micros...	4624	Logon
Information	5/17/2025 10:56:17 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:24 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:24 PM	Micros...	4732	Securit...
Information	5/17/2025 10:55:12 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:12 PM	Micros...	4732	Securit...
Information	5/17/2025 10:55:04 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:03 PM	Micros...	4798	User A...
Information	5/17/2025 10:54:59 PM	Micros...	4732	Securit...

Event 4732, Microsoft Windows security auditing.

General Details

A member was added to a security-enabled local group.

Subject:

Security ID: THM-PC\Administrator
Account Name: Administrator
Account Domain: THM-PC
Logon ID: 0x183C36D

Member:

Security ID: S-1-5-21-1966530601-3185510712-10604624-1013
Account Name: -

Group:

Security ID: BUILTIN\Remote Desktop Users
Group Name: Remote Desktop Users
Group Domain: Builtin

Additional Information:

Privileges: -

Practice-Security Number of events: 84

Information	5/17/2025 10:56:18 PM	Micros...	4624	Logon
Information	5/17/2025 10:56:17 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:24 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:24 PM	Micros...	4732	Securit...
Information	5/17/2025 10:55:12 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:12 PM	Micros...	4732	Securit...
Information	5/17/2025 10:55:04 PM	Micros...	4798	User A...
Information	5/17/2025 10:55:03 PM	Micros...	4798	User A...
Information	5/17/2025 10:54:59 PM	Micros...	4732	Securit...

Event 4732, Microsoft Windows security auditing.

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Subject:

Security ID: THM-PC\Administrator
Account Name: Administrator
Account Domain: THM-PC
Logon ID: 0x183C36D

Member:

Security ID: S-1-5-21-1966530601-3185510712-10604624-1013
Account Name: -

Group:

Security ID: BUILTIN\Backup Operators
Group Name: Backup Operators
Group Domain: Builtin

Additional Information:

Privileges: -

The attacker created a user and added two privileges: Backup Operators and Remote Desktop Users

- Alert on more than 5 failed logins (Event ID 4625) from the same source IP within 60 seconds.
- Alert on Event ID 4624 with Logon Type 10 (RDP) from an external or unfamiliar IP address.
- Alert on Sysmon Event 1 where a process is spawned by an unexpected parent (for example, cmd.exe spawned by explorer.exe shortly after a login).

Phase 2: Initial Access

Initial Access is the first foothold an attacker gains on a machine. The three main vectors investigated in this phase were RDP brute-force, phishing email attachments, and malicious USB execution.

RDP Brute-Force

The Administrator account was the most actively targeted by botnets across the captured logs. Filtering Event ID 4625 with Logon Type 10 isolated all failed RDP attempts. The exact IP that succeeded in breaching the host was then found in Event ID 4624, also with Logon Type 10.

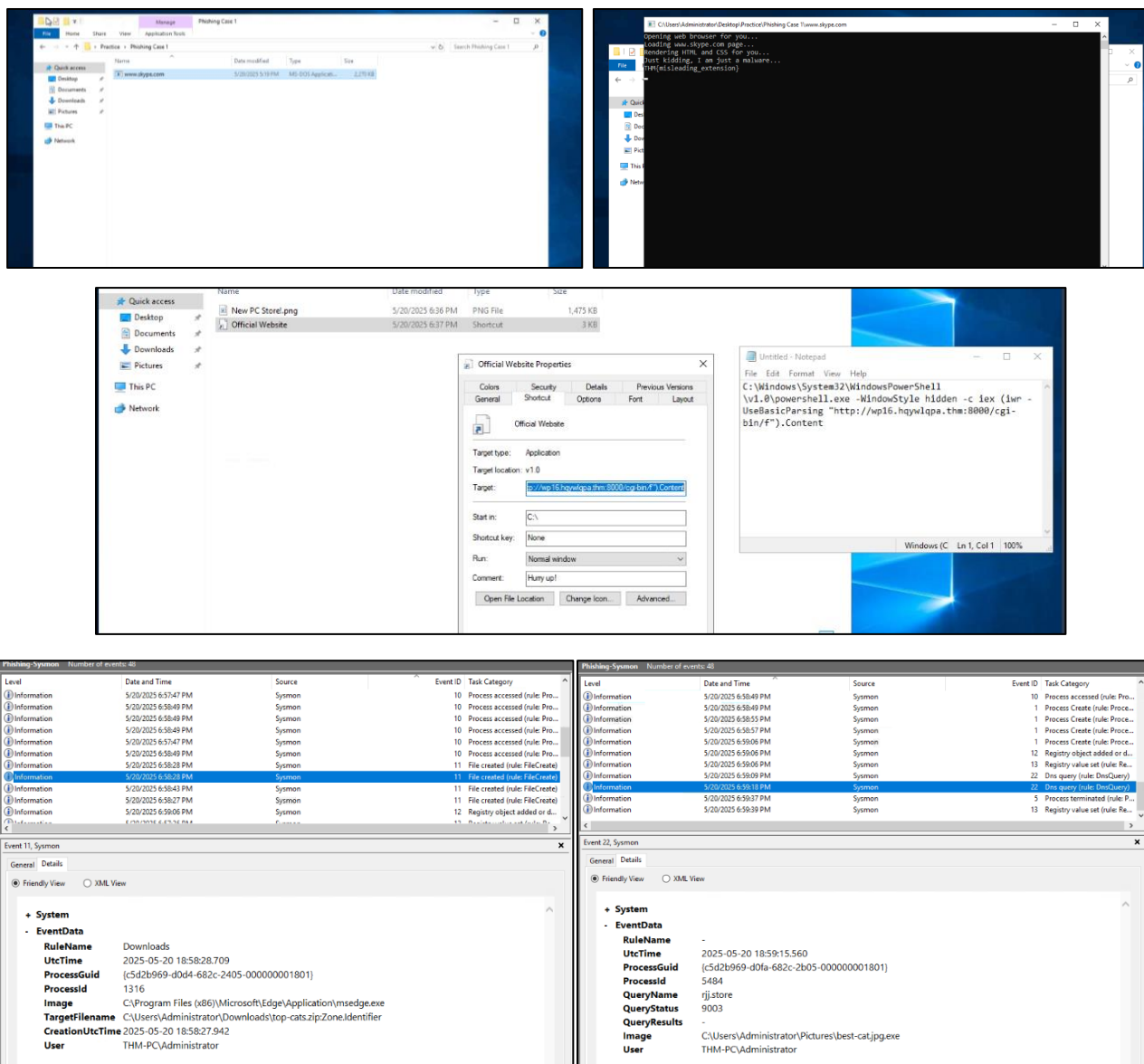
RDP-Security - Number of events: 1,567					RDP-Security - Number of events: 1,567				
Level	Date and Time	Source	Event ID	Task Category	Level	Date and Time	Source	Event ID	Task Category
Information	5/20/2025 9:51:49 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:49 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:48 AM	Microsoft Windows securi...	4625	Logon	Information	5/20/2025 9:51:48 AM	Microsoft Windows securi...	4625	Logon
Information	5/20/2025 9:51:48 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:48 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:48 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:48 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:47 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:47 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:47 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:47 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:47 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:47 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:46 AM	Microsoft Windows securi...	4625	Logon	Information	5/20/2025 9:51:46 AM	Microsoft Windows securi...	4625	Logon
Information	5/20/2025 9:51:45 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:45 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:45 AM	Microsoft Windows securi...	4624	Logon	Information	5/20/2025 9:51:45 AM	Microsoft Windows securi...	4624	Logon
Information	5/20/2025 9:51:43 AM	Microsoft Windows securi...	4625	Logon	Information	5/20/2025 9:51:43 AM	Microsoft Windows securi...	4625	Logon
Event 4625: Microsoft Windows security auditing.					Event 4624: Microsoft Windows security auditing.				
General Details					General Details				
An account failed to log on.					Logon ID: 0x0				
Subject:					Logon Information:				
Security ID: NULL SID					Logon Type: 3				
Account Name: -					Restricted Admin Mode: No				
Account Domain: -					Virtual Account: Yes				
Logon ID: 0x0					Elevated Token: Yes				
Logon Type: 3					Impersonation Level: Impersonation				
Account For Which Logon Failed:					New Logon:				
Security ID: NULL SID					Security ID: S-1-5-21-1592303528-961660316-3103943273-500				
Account Name: ADMINISTRATOR					Account Name: Administrator				
Account Domain: -					Account Domain: WIN-F907F8E10				
Failure Information:					Logon ID: 0x0				
Failure Reason: Unknown user name or bad password.					Linked Logon ID: 0x0				
Status: 0xC000006D					Network Account Name: -				
Sub Status: 0xC000006A					Network Account Domain: -				
Process Information:					Logon GUID: {00000000-0000-0000-0000-000000000000}				
Caller Process ID: 0x0					Process Information:				
Caller Process Name: -					Process ID: 0x0				
Network Information:					Process Name: -				
Workstation Name: -					Network Information:				
Source Network Address: 152.230.66.217					Workstation Name: DESKTOP-QNBC4UU				
Source Port: 0					Source Network Address: 203.205.34.107				
Detailed Authentication Information:					Source Port: 0				
Logon Process: NtLmSsp					Detailed Authentication Information:				
Authentication Package: NTLM					Logon Process: NtLmSsp				
Transited Services: -					Authentication Package: NTLM				
Package Name (NTLM only): -					Transited Services: -				
Key Length: 0					Package Name (NTLM only): NTLM V2				

Event log analysis revealed an active botnet brute-force campaign targeting a user account. The adversary, operating from workstation DESKTOP-QNBC4UU via IP address 203.205.34.107, successfully compromised the host.

Phishing Attachment Cases

Three phishing cases were investigated. In Case 1, a COM file disguised as a skype installer was executed by the user. Sysmon Event 1 showed explorer.exe as the parent process, confirming the user double-clicked the file through the Windows GUI.

In Case 2, a malicious LNK shortcut file silently downloaded a second-stage payload. The download URL was captured in Sysmon process creation logs as an argument passed to a command-line tool invoked by the LNK.



The user downloaded a malicious archive named top-cats.zip and extracted its contents into the "Pictures" directory. This action initiated a phishing malware process (PID: 5484), which subsequently attempted to establish outbound network connections to the malicious domain rjj[.]store.

USB Execution

Filtering Sysmon Event 1 for Image paths beginning with removable drive letters (D, E, F) revealed the file executed from the USB drive. A subsequent Sysmon Event 11 showed the secondary file dropped to the local disk by the initial payload.

A screenshot of the Windows Event Viewer application. The left-hand pane shows a tree view with 'System' expanded, and 'EventData' selected. The right-hand pane displays the details of a specific event. The details are organized into a table with two columns: a property name and its corresponding value. The properties listed are RuleName, EventType, UtcTime, ProcessGuid, ProcessId, Image, TargetObject, Details, and User. The values are: T1060.RunKey, SetValue, 2025-05-20 19:34:06.761, {c5d2b969-d92e-682c-9205-0000000001801}, 1108, E:\Open Sandisk 4GB USB\exe, HK\US-1.5-21-1966530601-3185510712-10604624-500\Software\Microsoft\Windows\CurrentVersion\Run\WinUpdate.exe, C:\Users\Public\Documents\winupdate.exe, and THM-PC\Administrator respectively.

The user executed a malicious binary from a removable drive located at E:\Open Sandisk 4GB USB.exe. Upon execution, the process dropped a suspicious executable named winupdate.exe into the C:\Users\Public\Documents\ directory.

DETECTION OPPORTUNITIES, PHASE 2

- Alert on Sysmon Event 1 where a process is executed from a drive letter associated with removable media (D, E, F).
- Alert on Sysmon Event 1 where a COM, LNK, or HTA file is executed with explorer.exe as the parent process.
- Alert on Event 4625 with Logon Type 10 exceeding a threshold from a single external IP within a short window.

Phase 3: Post-Breach Activity

After gaining access, attackers run discovery commands to understand the environment, collect credentials, and locate sensitive data. Sysmon Event 1 was the primary evidence source for all activity in this phase.

Execution and Discovery

Sysmon Event 1 logs every process with its full command line and parent process. Commands like net user Administrator, whoami, ipconfig, and systeminfo were identified in the logs by filtering for the relevant process names.

All events shared the same Logon ID from the attacker's RDP session, making it possible to confirm every command was executed by the same session and not by a legitimate user.

Operational Number of events: 2,674 (0) New events available					Operational Number of events: 2,674 (0) New events available				
Level	Date and Time	Source	Event ID	Task Category	Level	Date and Time	Source	Event ID	Task Category
Information	5/30/2026 12:28:08 PM	Sysmon	6	Driver loaded (rul...	Information	5/30/2026 12:28:01 PM	Sysmon	11	File created (rul...
Information	5/30/2026 12:28:06 PM	Sysmon	22	Dns query (rule...	Information	5/30/2026 12:28:01 PM	Sysmon	11	File created (rul...
Information	5/30/2026 12:28:04 PM	Sysmon	22	Dns query (rule...	Information	5/30/2026 12:28:01 PM	Sysmon	11	File created (rul...
Information	5/30/2026 12:28:04 PM	Sysmon	22	Dns query (rule...	Information	5/30/2026 12:28:01 PM	Sysmon	3	Network connec...
Information	5/30/2026 12:28:04 PM	Sysmon	22	Dns query (rule...	Information	5/30/2026 12:28:00 PM	Sysmon	1	Process Create (...)
Information	5/30/2026 12:28:04 PM	Sysmon	3	Network connec...	Information	5/30/2026 12:28:00 PM	Sysmon	1	Process Create (...)
Information	5/30/2026 12:28:04 PM	Sysmon	3	Network connec...	Information	5/30/2026 12:28:00 PM	Sysmon	11	File created (rul...
Information	5/30/2026 12:28:03 PM	Sysmon	3	Network connec...	Information	5/30/2026 12:28:00 PM	Sysmon	1	Process Create (...)
Information	5/30/2026 12:28:03 PM	Sysmon	11	File created (rul...	Information	5/30/2026 12:28:00 PM	Sysmon	1	Process Create (...)
Information	5/30/2026 12:28:03 PM	Sysmon	1	Process Create (...)	Information	5/30/2026 12:28:00 PM	Sysmon	1	Process Create (...)
Event 1, Sysmon					Event 1, Sysmon				
General Details					General Details				
Friendly View XML View					Friendly View XML View				
+ System					+ System				
- EventData					- EventData				
RuleName -					RuleName -				
UtcTime 2026-05-30 12:28:03.121					UtcTime 2026-05-30 12:28:00.037				
ProcessGuid {c5d2b969-d7d3-6a1a-b300-000000001d01}					ProcessGuid {c5d2b969-d7d0-6a1a-aa00-000000001d01}				
ProcessId 3432					ProcessId 996				
Image C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe					Image C:\Windows\System32\cmd.exe				
FileVersion 10.0.17763.1 (WinBuild.160101.0800)					FileVersion 10.0.17763.1697 (WinBuild.160101.0800)				
Description Windows PowerShell					Description Windows Command Processor				
Product Microsoft® Windows® Operating System					Product Microsoft® Windows® Operating System				
Company Microsoft Corporation					Company Microsoft Corporation				
OriginalFileName PowerShell.EXE					OriginalFileName Cmd.Exe				
CommandLine powershell -c "Compress-Archive -Force \$env:Temp\staging_58f1 - DestinationPath \$env:Temp\staging_58f1.zip"					CommandLine cmd /c "xcopy %UserProfile%\Downloads*.docx %Temp%\staging_58f1\downloads /i /s /y"				
CurrentDirectory C:\Users\Administrator\Desktop\Practice\Task 5\					CurrentDirectory C:\Users\Administrator\Desktop\Practice\Task 5\				
User THM-DFIR-VM-2\Administrator					User THM-DFIR-VM-2\Administrator				
LogonGuid {c5d2b969-d7a9-6a1a-3d6a-030000000000}					LogonGuid {c5d2b969-d7a9-6a1a-3d6a-030000000000}				
LogonId 0x36a3d					LogonId 0x36a3d				
TerminalSessionId 2					TerminalSessionId 2				
IntegrityLevel High					IntegrityLevel High				
Hashes MD5=7353F60B1739074E817C5F4DDDEF239.SHA256=DE96A6E69944335375DC1.					Hashes MD5=911D039E71583A07320B328DE22F8E22.SHA256=BC866CFDDA37E24DC263				
ParentProcessGuid {c5d2b969-d7cd-6a1a-9800-000000001d01}					ParentProcessGuid {c5d2b969-d7cd-6a1a-9800-000000001d01}				
ParentProcessId 3232					ParentProcessId 3232				
ParentImage C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe					ParentImage C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe				
ParentCommandLine "C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe"					ParentCommandLine "C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe"				
ParentUser THM-DFIR-VM-2\Administrator					ParentUser THM-DFIR-VM-2\Administrator				
Event 1, Sysmon					Event 22, Sysmon				
General Details					General Details				
Friendly View XML View					Friendly View XML View				
+ System					+ System				
- EventData					- EventData				
RuleName -					RuleName -				
UtcTime 2026-05-30 12:27:58.321					UtcTime 2026-05-30 12:28:05.699				
ProcessGuid {c5d2b969-d7ce-6a1a-9e00-000000001d01}					ProcessGuid {c5d2b969-d7cd-6a1a-9800-000000001d01}				
ProcessId 2672					ProcessId 3232				
Image C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe					QueryName collecteddata-storage-2025.s3.amazonaws.com				
FileVersion 10.0.17763.1 (WinBuild.160101.0800)					QueryStatus 0				
Description Windows PowerShell					QueryResults type: 5 s3-1-w.amazonaws.com;type: 5 s3-w-us-east-1.amazonaws.com;ffff52.217.207.49;ffff52.216.38.177;ffff16.15.254.123;ffff16.15.2				
Product Microsoft® Windows® Operating System					Image C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe				
Company Microsoft Corporation					User THM-DFIR-VM-2\Administrator				
OriginalFileName PowerShell.EXE									
CommandLine powershell -c "Get-Clipboard -> \$env:Temp\staging_58f1\clipboard.txt"									
CurrentDirectory C:\Users\Administrator\Desktop\Practice\Task 5\									
User THM-DFIR-VM-2\Administrator									
LogonGuid {c5d2b969-d7a9-6a1a-3d6a-030000000000}									
LogonId 0x36a3d									
TerminalSessionId 2									
IntegrityLevel High									
Hashes MD5=7353F60B1739074E817C5F4DDDEF239.SHA256=DE96A6E69944335375DC1.									
ParentProcessGuid {c5d2b969-d7cd-6a1a-9800-000000001d01}									
ParentProcessId 3232									
ParentImage C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe									
ParentCommandLine "C:\Users\Administrator\Desktop\Practice\Task 5\stealer.exe"									
ParentUser THM-DFIR-VM-2\Administrator									

A malicious file (stealer.exe) created a temporary folder named staging_58f1 to collect data. It specifically hunted for PDF, DOCX, and XLSX documents, used a PowerShell command to copy clipboard history, and uploaded all the stolen information to an AWS cloud bucket (collecteddata-storage-2025[.js[.amazonaws[.com]).

DETECTION OPPORTUNITIES, PHASE 3

- Alert on Sysmon Event 1 for discovery commands such as net user, whoami, ipconfig, or systeminfo executed outside of normal working hours or by an unexpected user account.
- Alert on Sysmon Event 1 where a browser process reads files from its credential storage directory, which may indicate automated credential dumping.

- Alert on Sysmon Event 11 where files with SSH key extensions (.pem, .ppk, id_rsa) are created or accessed by an unusual process.

Phase 4: C2, Persistence, and Impact

The final phase covers the most dangerous stage of a Windows attack: when the attacker establishes remote control, ensures they can always return, and executes their final objective.

Command and Control

The initial infection vector was a deceptive archive named URGENT!.zip, using social engineering to prompt the user to open it. The malicious executable inside established a C2 channel back to the attacker's infrastructure.

Sysmon Event 3 (network connection) was used to identify the outbound connection from the malware process. The callback domain was also captured in Sysmon DNS query logs.

Sysmon - Number of events: 23					Sysmon - Number of events: 23				
Level	Date and Time	Source	Event ID	Task Category	Level	Date and Time	Source	Event ID	Task Category
Information	7/2/2025 8:14:07 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:30 PM	Sysmon	22	Dns query (rule: DnsQa...
Information	7/2/2025 8:14:07 PM	Sysmon	11	File created (rule: FileCr...	Information	7/2/2025 8:14:28 PM	Sysmon	1	Process Create (rule: Pr...
Information	7/2/2025 8:13:58 PM	Sysmon	1	Process Create (rule: Pr...	Information	7/2/2025 8:14:14 PM	Sysmon	3	Network connection d...
Information	7/2/2025 8:13:55 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:13 PM	Sysmon	11	File created (rule: FileCr...
Information	7/2/2025 8:13:55 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:12 PM	Sysmon	1	Process Create (rule: Pr...
Information	7/2/2025 8:13:55 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:07 PM	Sysmon	15	File stream created (rul...
Information	7/2/2025 8:13:55 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:07 PM	Sysmon	15	File stream created (rul...
Information	7/2/2025 8:13:55 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:07 PM	Sysmon	15	File stream created (rul...
Information	7/2/2025 8:13:55 PM	Sysmon	11	File created (rule: FileCr...	Information	7/2/2025 8:14:07 PM	Sysmon	11	File created (rule: FileCr...
Information	7/2/2025 8:13:50 PM	Sysmon	15	File stream created (rul...	Information	7/2/2025 8:14:07 PM	Sysmon	15	File stream created (rul...
Information	7/2/2025 8:13:38 PM	Sysmon	11	File created (rule: FileCr...	Information	7/2/2025 8:14:07 PM	Sysmon	11	File created (rule: FileCr...
Information	7/2/2025 8:13:38 PM	Sysmon	1	Process Create (rule: Pr...	Information	7/2/2025 8:14:07 PM	Sysmon	11	File created (rule: FileCr...

Event 11, Sysmon

General

Details

Friendly View

XML View

+ System

- EventData

RuleNameDownloads

UtcTime2025-07-02 20:13:55.860

ProcessGuid{c5d2b969-9303-6865-d300-000000001b01}

ProcessId5708

ImageC:\Program Files (x86)\Google\Chrome\Application\chrome.exe

TargetFilenameC:\Users\Administrator\Downloads\URGENT!.zipZone.Identifier

CreationUtcTime2025-07-02 20:13:50.234

UserTHM-DFIR-VM-3\Administrator

Event 22, Sysmon

General

Details

Friendly View

XML View

+ System

- EventData

RuleName-

UtcTime2025-07-02 20:14:28.636

ProcessGuid{c5d2b969-9324-6865-d700-000000001b01}

ProcessId4340

QueryNameroute.m365officesync.workers.dev

QueryStatus9003

QueryResults-

ImageC:\Users\Administrator\AppData\Roaming\update.exe

UserTHM-DFIR-VM-3\Administrator

The incident began when a user downloaded a suspicious archive titled URGENT!.zip. Upon opening the file, a hidden malicious program was installed in the user's system folders (C:/Users/Administrator/Appdata/Roaming/update.exe). This program was configured to communicate with an external command server to receive further instructions from the attacker.

Persistence Mechanisms

Three persistence methods were identified and documented.

- Backdoor user account: Event ID 4720 showed a new local account was created, immediately followed by Event ID 4728 adding it to the Administrators group.
- Registry Run Key: Sysmon Event 13 (registry value set) captured a new entry added to the Windows Run key, causing the malware to launch on every system boot.
- Scheduled Task: Event ID 4698 recorded the creation of a task named AmazonSync, disguised to blend in with legitimate cloud service activity.

Security - Number of events: 31

Filtered Log file://C:/Users/Administrator/Desktop/Practice/Task 3/Security.evtx; Source : ; Event ID: 4625. Number of events: 6

Level	Date and Time	Source	Event ID	Task Category
Information	7/2/2025 9:00:20 PM	Microsoft Windows se...	4625	Login
Information	7/2/2025 9:58:22 PM	Microsoft Windows se...	4625	Login
Information	7/2/2025 9:58:18 PM	Microsoft Windows se...	4625	Login
Information	7/2/2025 9:58:13 PM	Microsoft Windows se...	4625	Login
Information	7/2/2025 9:58:09 PM	Microsoft Windows se...	4625	Login
Information	7/2/2025 9:58:05 PM	Microsoft Windows se...	4625	Login

Event 4625, Microsoft Windows security auditing.

General

Details

An account failed to log on.

Subject:

Security ID: NULL SID
Account Name: -
Account Domain: -
Logon ID: 0x0

Logon Type: 3

Account For Which Logon Failed:

Security ID: NULL SID
Account Name: Administrator
Account Domain: DESKTOP-4KCMRT

Failure Information:

Failure Reason: Unknown user name or bad password.
Status: 0xC000006D
Sub Status: 0xC000006A

Process Information:

Caller Process ID: 0x0
Caller Process Name: -

Network Information:

Workstation Name: DESKTOP-4KCMRT
Source Network Address: 10.14.97.15
Source Port: 0

Detailed Authentication Information:

Logon Process: NtLmSsp
Authentication Package: NTLM
Transmitted Services: -
Package Name (NTLM only): -
Key Length: 0

Security - Number of events: 31

Level	Date and Time	Source	Event ID	Task Category
Information	7/2/2025 9:01:38 PM	Microsoft Windows se...	4738	User Account Manage...
Information	7/2/2025 9:01:38 PM	Microsoft Windows se...	4722	User Account Manage...
Information	7/2/2025 9:01:38 PM	Microsoft Windows se...	4720	User Account Manage...

Event 4720, Microsoft Windows security auditing.

General

Details

A user account was created.

Subject:

Security ID: THM-DFR-VM-3\Administrator
Account Name: Administrator
Account Domain: THM-DFR-VM-3
Logon ID: 0x1CBC41

New Account:

Security ID: THM-DFR-VM-3\support
Account Name: support
Account Domain: THM-DFR-VM-3

Attributes:

SAM Account Name: support
Display Name: <value not set>
User Principal Name: -
Home Directory: <value not set>
Home Drive: <value not set>
Script Path: <value not set>
Profile Path: <value not set>
User Workstations: <value not set>
Password Last Set: <never>
Account Expires: <never>
Primary Group ID: 513
Allowed To Delegate To: -
Old UAC Value: 0x0
New UAC Value: 0x15
User Account Control: Account Disabled
Password Not Required - Enabled
Normal Account - Enabled
User Parameters: <value not set>
SID History: -
Logon Hours: All

Additional Information:

Privileges: -

System logs indicate a targeted login attempt against the Administrator account. After six failed attempts, the attacker successfully gained access on the seventh try. Once inside, they ensured future access by creating a new backdoor user account named support and elevating it to the Administrators group.

Security (All Events) - Number of events: 31

Level

Date and Time

Source

Event ID

Task Category

Information

7/2/2025 10:00:20 PM

Microsoft Windows se...

4624

Login

Information

7/2/2025 10:00:34 PM

Microsoft Windows se...

4624

Login

Information

7/2/2025 10:00:34 PM

Microsoft Windows se...

4627

Security System Emergen...

Information

7/2/2025 10:00:34 PM

Microsoft Windows se...

4627

Security System Emergen...

Event 4627, Microsoft Windows security auditing

General

Details

Friendly View

XML View

System

EventData

SubjectUserSid

S-1-5-21-1966536061-3185510712-10604624-500

SubjectUserName

Administrator

SubjectDomainName

THM-DFR-VM-3

SubjectLogonId

0x3d349

ServiceName

Data Protection Service

ServiceFileName

C:\Windows\Help\psess.exe

ServiceType

0x10

ServiceStartType

2

ServiceAccount

LocalSystem

ClientProcessStartKey

7599446845694824

ClientProcessId

5932

ParentProcessId

6616

Security (All Events) - Number of events: 31

Level

Date and Time

Source

Event ID

Task Category

Information

7/2/2025 10:02:30 PM

Microsoft Windows se...

4738

User Account Manage...

Information

7/2/2025 10:00:34 PM

Microsoft Windows se...

4624

Login

Information

7/2/2025 10:00:34 PM

Microsoft Windows se...

4627

Security System Emergen...

Event 4627, Microsoft Windows security auditing

General

Details

Friendly View

XML View

System

EventData

SubjectUserSid

S-1-5-21-1966536061-3185510712-10604624-500

SubjectUserName

Administrator

SubjectDomainName

THM-DFR-VM-3

SubjectLogonId

0x3d349

TaskName

Amazonsync

TaskContent

<?xml version="1.0" encoding="UTF-16"?><Task version="1.2" xmlns="http://schemas.microsoft.com/windows/2004/02/mi/task"><RegistrationInfo><Date>2025-07-02T19:10:09.000Z</Date><Author>THM-DFR-VM-3\Administrator</Author><AuthorUserSid>S-1-5-21-1966536061-3185510712-10604624-500</AuthorUserSid><IsHidden>false</IsHidden><Enabled>true</Enabled><BootTrigger><Triggers><Settings><MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy><DisallowStartOnRemoval>true</DisallowStartOnRemoval><DisallowStopOnRemoval>true</DisallowStopOnRemoval></Settings></RegistrationInfo><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><Triggers><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To maintain hidden access, the attacker established a fraudulent Windows service named "Data Protection Service" alongside a fake scheduled task called "Amazonsync," which was designed to masquerade as a legitimate Amazon utility. Once active, this trojan malware ensures its persistence by executing its background processes through the standard system using the command C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule.

DETECTION OPPORTUNITIES, PHASE 4

- Alert on Event 4720 (user created) followed within 60 seconds by Event 4728 (added to Administrators) for the same account name.
- Alert on Event 4698 (scheduled task created) where the task action path points to a non-standard directory such as Temp, AppData, or ProgramData.
- Alert on Sysmon Event 13 (registry set) for the Run or RunOnce registry keys by any process other than a known software installer.
- Alert on Sysmon Event 3 where a process with no legitimate network function initiates an outbound connection to an external IP on an unusual port.

Indicators of Compromise (IOCs)

The following indicators were identified across all four phases of the investigation.

IOC Type	Value	Context
IP Address	193.46.217.4	Adversary IP. Source of C2 reverse shell connection on port 7777 and web enumeration activity.
IP Address	10.10.133.168	Internal VPN-segment device used to access the GitLab instance during initial intrusion.
Username	stansimon	Target of brute-force SSH authentication attempts recorded in rsyslog_sshd.log.
C2 Endpoint	193.46.217.4:7777	Destination of the reverse bash shell executed via a compromised root cron job.
Malware File	URGENT!.zip	Deceptive archive used as the initial infection vector. Downloaded by a user on the target machine.
Scheduled Task	AmazonSync	Fake scheduled task created for persistence. Disguised with a cloud service name to avoid suspicion.
Credential	nsAghv51BBav90!	Facebook password recovered from Chrome Password Manager during post-breach credential collection.
Phishing File	www.skype.com (COM file)	Malicious COM file disguised as a skype installer. Executed by the user via Windows Explorer.

MITRE ATT&CK Technique Mapping

Phase	Technique ID	Technique Name	Observed In
Initial Access	T1133	External Remote Services (RDP)	Phase 2, RDP brute-force
Initial Access	T1566	Phishing (Email Attachment)	Phase 2, COM and LNK files
Initial Access	T1091	Replication via Removable Media	Phase 2, USB execution
Discovery	T1087	Account Discovery (net user)	Phase 3, post-breach enumeration
Credential Access	T1555.003	Credentials from Web Browsers	Phase 3, Chrome saved passwords
C2	T1071.001	Application Layer Protocol C2	Phase 4, malware callback domain
Persistence	T1136	Create Account	Phase 4, backdoor admin user

Persistence	T1547.001	Registry Run Keys	Phase 4, registry Run key
Persistence	T1053.005	Scheduled Task	Phase 4, AmazonSync task
Impact	T1486	Data Encrypted for Impact	Phase 4, Kitten ransomware

Project Summary

This four-phase project traced a complete Windows attack chain from first contact to full system compromise. An attacker brute-forces an exposed RDP port and gains access. They run discovery commands, harvest credentials from the browser and disk, and locate sensitive files. They deploy malware via a deceptive archive that establishes a C2 channel. They then create a backdoor admin account, add a registry Run key, and disguise a scheduled task as a cloud service. Finally, they execute ransomware across the system. Every step was visible in Windows logs. The analyst's job is to know which Event IDs to filter for, how to correlate events using shared identifiers, and how to read the story those events tell before the attacker finishes it.

Area Covered	Key Takeaway
Phase 1, Logging	Established log sources and Event IDs. Completed brute-force and RDP session investigation using Event Viewer and Sysmon correlation.
Phase 2, Initial Access	Detected RDP brute-force (T1133), phishing payloads (T1566), and USB execution (T1091) using Security and Sysmon logs.
Phase 3, post-Breach	Traced discovery commands, browser credential harvesting (T1555.003), SSH key on disk, and sensitive file access via Sysmon process logs.
Phase 4, C2 and Persistence	Detected C2 callback, backdoor account, registry Run key, and AmazonSync scheduled task. Identified ransomware impact stage.
Tools Used	Windows Event Viewer, Sysmon, Security .evtx logs, PowerShell logs, Logon ID correlation.
IOCs Identified	Adversary IP 193[.]46[.]217[.]4, C2 port 7777, URGENT!.zip, AmazonSync task, phishing COM file, Chrome credential nsAghv51BBav90!.